

Robotics & AI Club

Phase 2 Curriculum Plan

Classes 17-32 · 16 new classes · 2 a week · 8 weeks

4 age pathways · 64 new activities · about half coding · capstone showcase

INTERNAL · FOR ACADEMIC LEADERSHIP



Little Explorers

Ages 5-7

Think, play, code and invent



Junior Makers

Ages 8-11

Code it, light it, build it



Young Innovators

Ages 12-14

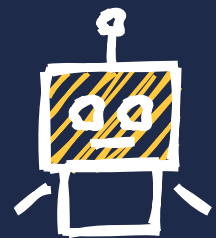
Creative code meets real hardware



Tech Leaders

Ages 15-17

Vision, algorithms and real products



CURRICULUM DEVELOPED BY

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4

age pathways



16

classes per batch (2 a week, 8 weeks)



+2

optional capstone sessions after the batch



2

students per kit group

How every class runs (90-120 min)



1. Hook

A real-world question, video or demo (10 min)



2. Guided build

Step-by-step with the teacher; simulate first where possible



3. Discovery

Open challenge: change, extend or improve



4. Share

Show, explain and reflect (5-10 min)

Pathways and progression

Pathway	Classes 17-20 (Weeks 1-2)	Classes 21-24 (Weeks 3-4)	Classes 25-28 (Weeks 5-6)	Classes 29-32 (Weeks 7-8)	Mix (H/S/HS/U)
Little Explorers Ages 5-7	Think Like a Computer Unplugged algorithms, Code.org puzzles, a programmed robot car and a ScratchJr dance party	Light, Sound & Power A rainbow LED ring, a button melody maker, a colour detective and a crank-powered light	Stories, Pictures & Machines A talking Scratch pet, pixel messages, a camera-controlled game and a cardboard automaton	Smart Helpers & AI Teaching a computer sounds, a follow-me robot pet, safety helper robots and a rocket launch	5 / 5 / 3 / 3
Junior Makers Ages 8-11	Code Studio Python Turtle art, a MakeCode Arcade game, binary secret codes and a joystick maze	Light, Sound & Music Tech NeoPixel light shows, an LED matrix, an air theremin and Processing art	Smart City & Mechanisms A smart traffic junction, a kitchen scale, a claw machine and wind turbines	Code, Data & AI A Python guessing game, class data detectives, an AI emotion mirror and a cyber escape room	6 / 6 / 2 / 2
Young Innovators Ages 12-14	Creative Coding Processing art and radar, a Pygame game and an ESP32-CAM wildlife camera	Data & Signals Python data logging, a posture wearable, Li-Fi messages and code breaking	Robots & Mechanisms Robot sumo, a walking linkage robot, an energy meter and a Python control panel	AI & the Web A voice-controlled p5.js game, a weather API dashboard, sign-language AI and a custom controller	6 / 7 / 3 / 0
Tech Leaders Ages 15-17	Vision & Intelligence OpenCV tracking, a vision-guided car, TinyML gestures and flocking simulations	Algorithms & Systems Pathfinding algorithms, Git teamwork, a solar microgrid and noise analysis with FFT	Robots & Control Arm inverse kinematics, a robot swarm, a 3D-printed gearbox and a smart cane	AI, Data & Product Pakistan air-quality data science, a neural network from scratch, Godot games and password security	3 / 9 / 4 / 0

Design principles

- All 64 activities are new: none repeats Phase 1, and every pathway still covers all 10 domains.
- Simulate first (Tinkercad Circuits, Wokwi, PhET, Falstad), then build on real hardware.
- About half of each pathway is coding: Code.org and Scratch (5-7), Python, MakeCode and Processing (8-11), Processing, Python, p5.js (12-14), OpenCV, Godot, Git and ML (15-17).
- Ages 5-7 use pre-wired Arduino boards and Makey Makey; ages 8+ wire their own circuits.

Safety and policy notes

- Ages 5-7: the teacher pre-wires every circuit; children plug, press and program with blocks.
- Flame and gas tests are teacher-only; relays switch low-voltage loads only.
- 18650 cells (TL-23): protected cells, supervised charging only. LED strips run at low brightness on USB power.
- Cameras (LE-27, JM-31, YI-20, YI-31, TL-17/18): no recording of people without consent; GitHub accounts are 13+ only.

Phase 2 is the second 16-class batch. It keeps the same class format, age pathways and safety rules, but every activity is new and roughly half of each pathway is **coding-focused**: creative coding in Processing and p5.js, Python for games, data and hardware control, block coding for the youngest, and professional tools such as OpenCV, Git and Godot for the oldest.

Little Explorers

Ages 5-7 · LE-17 to LE-32

8 / 16

coding-focused classes

Code.org · ScratchJr · Scratch 3 (video sensing) · mBlock 5 live mode · Teachable Machine

Junior Makers

Ages 8-11 · JM-17 to JM-32

8 / 16

coding-focused classes

Python Turtle · MakeCode Arcade · Processing 4 · CODAP · PictoBlox AI · ArduinoBlocks

Young Innovators

Ages 12-14 · YI-17 to YI-32

10 / 16

coding-focused classes

Processing 4 · Python + Pygame · Python data + APIs · p5.js (JavaScript) · Tkinter + pyserial · ESP32-CAM

Tech Leaders

Ages 15-17 · TL-17 to TL-32

9 / 16

coding-focused classes

OpenCV · Edge Impulse (TinyML) · Processing · NumPy + pandas · Godot Engine · Git + GitHub

Who can join Phase 2?

- Ideal for students who completed Phase 1 in the same pathway.
- New students can join: teach the Foundation decks F-00 to F-03 (robotics, Arduino, ESP32, 3D printing) in the first week.
- Text-coding classes give printed starter code so nobody starts from a blank screen.

Before the batch starts

- Install Processing 4, Python (Thonny) with pygame, pyserial, matplotlib, requests and OpenCV on lab laptops.
- Create teacher class accounts: Code.org, Scratch, Tinkercad, Edge Impulse; GitHub for ages 13+ only.
- Order the new parts on the "New hardware" page; assemble and test the 4WD cars.

New software introduced in Phase 2

Tool	What it is	Ages	Used in
Code.org	Free block-coding puzzle courses	5-8	LE-18
MakeCode Arcade	Microsoft's free block/JavaScript game maker	8-14	JM-18
Processing 4	Free creative-coding language for visual art and interaction	10-17	JM-24, YI-17, YI-18, TL-20
p5.js web editor	Processing for the web, in JavaScript	12-17	YI-29, YI-31
Python + Pygame / Tkinter	Python libraries for games and desktop apps	10-17	JM-17, JM-29, YI-19, YI-28, YI-32, TL-21
CODAP	Free data-exploration tool for schools	8-14	JM-30
PictoBlox	Scratch-based coding with machine-learning extensions	8-14	JM-31
OpenCV / NumPy / pandas	Python libraries for vision, maths and data	15-17	TL-17, TL-18, TL-24, TL-29, TL-30
Edge Impulse	Free TinyML platform for microcontrollers	15-17	TL-19
Godot Engine	Free open-source game engine (GDScript)	14-17	TL-31
Git + GitHub	Version control and team code hosting (13+)	13-17	TL-22

ISTE Standards for Students
International Society for Technology in Education
Empowered learners, computational thinkers, innovative designers, digital citizens

CSTA PK-12 CS Standards (2026)
Computer Science Teachers Association
Algorithms, computing systems, data, networks and impacts, now with AI

NGSS Engineering Design
Next Generation Science Standards
Define a problem, design solutions, test and improve

UNESCO AI Competency Framework
UNESCO, 2024
Understand, apply and create with AI, with a human-centred and ethical mindset

Gold Standard PBL
PBLWorks
Real problems, student voice and choice, critique, revision and a public product

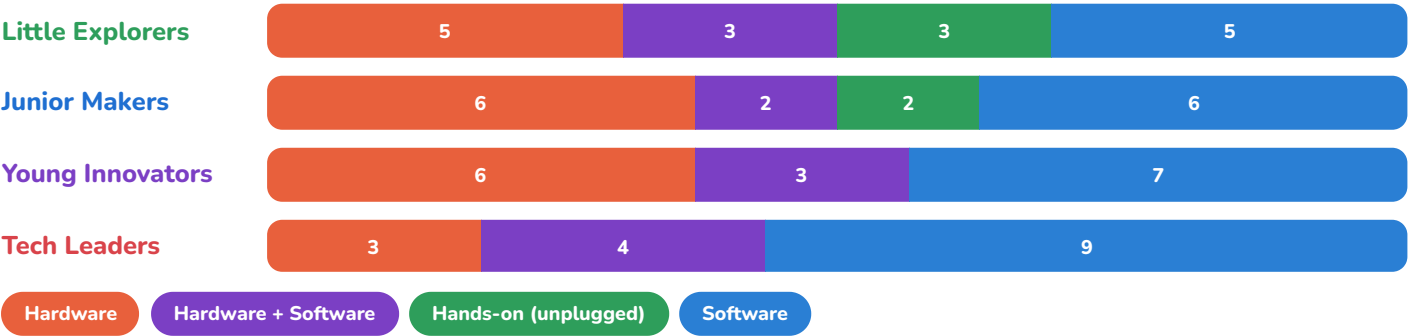
Plus UN SDGs
Every activity also carries UN Sustainable Development Goal badges (next page).

Domain coverage (number of classes touching each domain)

Domain	Little Explorers Ages 5-7	Junior Makers Ages 8-11	Young Innovators Ages 12-14	Tech Leaders Ages 15-17
Electronics & Circuits	2	3	3	1
Sensors & Environment	3	4	4	2
Robotics & Actuation	2	1	1	3
Mechanical Design	2	1	1	2
Power & Energy	1	1	1	1
Programming & Logic	6	4	2	4
IoT & Communication	1	1	3	3
Security & Safety	1	2	1	1
AI & Data	1	3	2	5
App, Web & Game Development	4	3	6	2

Counts include each activity's main and second domain. Every pathway covers all 10 domains.

Hands-on vs screen time



Each activity is tagged with the UN Sustainable Development Goals (SDGs) it helps students understand. Badges appear on every activity page, like achievements in a game, so students see how their builds connect to real global challenges.



SDG	Little Explorers Ages 5-7	Junior Makers Ages 8-11	Young Innovators Ages 12-14	Tech Leaders Ages 15-17	Example activities
SDG 2 · Zero Hunger	-	1	-	-	Smart Kitchen Scale
SDG 3 · Good Health	2	1	2	3	Pop the Bubbles with Your Hand; Safety Helper Robot Design
SDG 4 · Quality Education	11	12	4	2	Robot Chef: Unplugged Algorithms; Code.org Puzzle Quest: Sequences and Loops
SDG 7 · Clean Energy	2	2	1	1	Rainbow LED Ring; Crank-Powered Light: Be the Power Station
SDG 8 · Decent Work	-	-	-	2	Git & GitHub for Engineering Teams; 2D Game Development in Godot
SDG 9 · Industry & Innovation	7	3	9	11	Program the Robot Car Through a Floor Maze; Colour Detective Sensor
SDG 10 · Reduced Inequalities	-	1	2	1	AI Emotion Mirror; Sign Language Alphabet Recogniser
SDG 11 · Sustainable Cities	1	1	1	4	Safety Helper Robot Design; Smart Traffic Junction with Car Counter
SDG 12 · Responsible Consumption	1	1	1	-	Colour Detective Sensor; Smart Kitchen Scale
SDG 13 · Climate Action	1	1	2	2	Crank-Powered Light: Be the Power Station; Wind Turbine Power Lab
SDG 15 · Life on Land	-	-	1	1	ESP32-CAM Wildlife Camera; Processing: Flocking Simulation (Boids)
SDG 16 · Peace & Justice	-	2	1	1	Binary Bracelets & Secret Codes; Cyber Safety Escape Room
SDG 17 · Partnerships	1	-	-	-	Pixel Messages: Send a Picture with Numbers

Badges are an internal teaching aid. Numbers and goal names follow the United Nations 2030 Agenda; badge artwork is our own.



Phase 2 roadmap: classes 17-32

8 weeks, 2 classes a week: four phases of 4 classes. Builds on Phase 1; the capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Think Like a Computer

Unplugged algorithms, Code.org puzzles, a programmed robot car and a ScratchJr dance party

C17



Robot Chef: Unplugged Algorithms

Programming & Logic • **Hands-on (unplugged)**

Algorithms are exact step-by-step instructions

LE-17

C18



Code.org Puzzle Quest: Sequences and Loops

Programming & Logic • **Software**

Sequences and loops

LE-18

C19



Program the Robot Car Through a Floor Maze

Robotics & Actuation • **Hardware + Software**

Motors and timing make a robot move

LE-19

C20



ScratchJr Dance Party

App, Web & Game Development • **Software**

Events and loops

LE-20

Phase 2

Weeks 3-4

Light, Sound & Power

A rainbow LED ring, a button melody maker, a colour detective and a crank-powered light

C21



Rainbow LED Ring

Electronics & Circuits • **Hardware**

Mixing coloured light

LE-21

C22



Button Melody Maker

Electronics & Circuits • **Hardware**

Sound is made by vibrations

LE-22

C23



Colour Detective Sensor

Sensors & Environment • **Hardware**

Light and colour

LE-23

C24



Crank-Powered Light: Be the Power Station

Power & Energy • **Hardware**

Movement can be turned into electricity

LE-24

Phase 3

Weeks 5-6

Stories, Pictures & Machines

A talking Scratch pet, pixel messages, a camera-controlled game and a cardboard automaton

C25



My Talking Pet in Scratch

App, Web & Game Development • **Software**

Events, sounds and costumes

LE-25

C26



Pixel Messages: Send a Picture with Numbers

IoT & Communication • **Hands-on (unplugged)**

Pictures are stored as numbers (pixels)

LE-26

C27



Pop the Bubbles with Your Hand

App, Web & Game Development • **Software**

Camera input and variables

LE-27

C28



Cardboard Automaton: Cams and Cranks

Mechanical Design • **Hardware**

Cams change turning into up-and-down motion

LE-28

Phase 4

Weeks 7-8

Smart Helpers & AI

Teaching a computer sounds, a follow-me robot pet, safety helper robots and a rocket launch

C29



Teach the Computer Our Sounds

AI & Data • **Software**

Computers learn from examples

LE-29

C30



Follow-Me Robot Pet

Robotics & Actuation • **Hardware + Software**

Sensors guide motors

LE-30

C31



Safety Helper Robot Design

Security & Safety • **Hands-on (unplugged)**

Staying safe at home and school

LE-31

C32



Countdown Rocket Launch

App, Web & Game Development • **Hardware + Software**

Counting variables and hardware events

LE-32



After class 32 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: A robot helper for our classroom; A light-and-sound greeting machine; An interactive Scratch story with a real button; A cardboard automaton that tells a story. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
17**Robot Chef: Unplugged Algorithms**

Programming & Logic • Hands-on (unplugged) • Week 1

LE-17

Introduction

Computers only do exactly what they are told, step by step. In this no-screen class one child becomes the 'robot chef' and the others write arrow and picture cards to guide it through making a pretend sandwich or crossing a floor grid. When the robot does something silly, the team finds the wrong card and fixes it. Children discover algorithms, sequences and debugging before touching a computer.

**Software****No software needed**

Also: Code.org unplugged lessons

Key hardware (per group)

- Picture cards and stationery

Standards

CSTA Algorithms & Programming (1A) • ISTE Computational Thinker

SDG badges

Activities

GUIDED

- Write a picture-card algorithm to move a friend across a floor grid
- Find and fix the 'bug' when the robot goes the wrong way

DISCOVERY

- Use a 'repeat 3 times' card to make the program shorter

Outcomes

Computing: Algorithms are exact step-by-step instructions**Life Skills:** Giving clear instructions and listening**Maths:** Directions, counting and positionCLASS
18**Code.org Puzzle Quest: Sequences and Loops**

Programming & Logic • Software • Week 1

LE-18

Introduction

Code.org puzzles turn coding into a game: children drag picture blocks to guide a character to a goal. They start with simple sequences, then discover that a loop can replace many repeated blocks. Each finished level gives instant feedback, so children try, test and improve on their own. The class celebrates the shortest working solutions.

**Software****Code.org (Course A/B, teacher class code)**

Also: Lightbot Jr

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1A) • ISTE Computational Thinker

SDG badges

Activities

GUIDED

- Solve sequence puzzles by dragging arrow blocks
- Use a repeat loop to finish a puzzle with fewer blocks

DISCOVERY

- Help a partner find the shortest solution to a level

Outcomes

Computing: Sequences and loops**Life Skills:** Persistence when a puzzle is hard**Maths:** Counting steps and patterns

CLASS
19

Program the Robot Car Through a Floor Maze

Robotics & Actuation + Programming & Logic · Hardware + Software · Week 2

LE-19

Introduction	Delivery robots in warehouses follow programmed routes. Children use picture blocks to drive a pre-built 4-wheel robot car through a taped maze on the floor: forward for 1 second, turn, forward again. They measure the maze with their feet, plan the route on paper and adjust the timings when the car overshoots. This links their unplugged algorithms to a real moving robot.	 Software mBlock 5 (picture blocks, teacher-uploaded) Key hardware (per group) 4WD robot car (assembled, L298N) x1 Standards CSTA Computing Systems (1A) · NGSS K-2-ETS1 SDG badges 9 Industry & Innovation 4 Quality Education
Activities	GUIDED - Plan a route on a paper map of the maze - Program forward and turn blocks and test the car in the maze DISCOVERY - Add a light or a beep when the car reaches the finish	
Outcomes	Engineering: Motors and timing make a robot move Computing: Sequences on a real robot Maths: Measuring and estimating distance	

CLASS
20

ScratchJr Dance Party

App, Web & Game Development · Software · Week 2

LE-20

Introduction	Animators and game designers make characters move to music. Children create a dance party in ScratchJr: they choose characters, record or pick sounds and program dance moves that repeat in a loop. Two characters dance together when the green flag is tapped. They learn that programs start with an event and that loops make animations go on and on.	 Software ScratchJr Also: Scratch 3 (teacher demo) Key hardware (per group) Tablet (1 per group) Standards CSTA Algorithms & Programming (1A) · ISTE Creative Communicator SDG badges 4 Quality Education
Activities	GUIDED - Make a character dance with move and turn blocks - Start two dancers at the same time with the green flag DISCOVERY - Add a background change or a new dancer that joins later	
Outcomes	Computing: Events and loops Arts: Music, rhythm and animation Life Skills: Presenting your work to friends	

CLASS

21

Rainbow LED Ring

Electronics & Circuits • Hardware • Week 3

LE-21

Introduction	Smart lights can show millions of colours using tiny red, green and blue lights. Children program a pre-wired ring of 12 colour LEDs: they choose colours, light the ring one LED at a time and make a spinning rainbow. They discover that mixing red, green and blue light makes new colours. The ring becomes a magic lamp they can show at home time.	 Software ArduinoBlocks (teacher-led, LED ring blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • WS2812B LED ring (12 LEDs) x1 Standards NGSS K-2-ETS1 • CSTA Algorithms & Programming (1A) SDG badges 7 Clean Energy 4 Quality Education
Activities	GUIDED • Pick colours for the ring and light it all at once • Make one light chase around the ring with a loop DISCOVERY • Invent a 'mood' colour for happy, sleepy and excited	
Outcomes	Science: Mixing coloured light Computing: Loops make patterns Arts: Designing with colour	

CLASS

22

Button Melody Maker

Electronics & Circuits + Programming & Logic • Hardware • Week 3

LE-22

Introduction	Electronic pianos and toys play notes when you press a key. Children use three pre-wired buttons and a small speaker (a passive buzzer) to play notes, then use picture blocks to choose which note each button plays. They play and record a short tune, then make a 'surprise' button that plays a whole melody. Music becomes their first taste of inputs and outputs working together.	 Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • Push button x3 • Passive buzzer x1 • Resistor 10k ohm x3 Standards NGSS 1-PS4 Waves (sound) • CSTA Computing Systems (1A) SDG badges 4 Quality Education
Activities	GUIDED • Make each button play a different note • Program one button to play a short tune DISCOVERY • Compose and perform a 4-note tune for the class	
Outcomes	Science: Sound is made by vibrations Computing: Inputs trigger outputs Arts: Composing a simple tune	

CLASS
23

Colour Detective Sensor

Sensors & Environment • Hardware • Week 4

LE-23

Introduction

Factories use colour sensors to sort sweets, fruit and recycling. Children hold coloured cards over a colour sensor and watch an RGB light copy the colour it sees. They play 'guess the colour' and test tricky colours like brown, pink and white. They discover that machines see colour as a mix of red, green and blue numbers.

Activities

GUIDED

- Show coloured cards to the sensor and watch the light copy them
- Find a colour that confuses the sensor

DISCOVERY

- Sort a pile of cards into red, green and blue using the machine

Outcomes

Science: Light and colour

Technology: Sensors in factories

Computing: Machines turn colour into numbers



Software

Teacher-uploaded program (Arduino IDE 2)

Key hardware (per group)

- Arduino Uno + USB cable x1
- Colour sensor TCS34725 x1
- RGB LED x1
- Resistor 220 ohm x3

Standards

NGSS K-2-ETS1 • CSTA Data & Analysis (1A)

SDG badges


CLASS
24

Crank-Powered Light: Be the Power Station

Power & Energy • Hardware • Week 4

LE-24

Introduction

Power stations make electricity by spinning a generator. Children discover that a small motor works backwards as a generator: when they spin it fast with a crank or a pulley, an LED flickers on and a multimeter shows the voltage rise. They race to keep the light on as long as possible and talk about where our electricity comes from. Children feel that electricity needs energy from somewhere.

Activities

GUIDED

- Spin the motor and watch the LED and the meter
- Compare slow and fast spinning

DISCOVERY

- Make a paper windmill and test if wind can turn the motor

Outcomes

Science: Movement can be turned into electricity

Technology: Power stations and wind turbines

Life Skills: Saving electricity at home



Software

No software needed

Also: PhET (teacher demo)

Key hardware (per group)

- DC hobby motor x1
- LED 5 mm (assorted) x2

Standards

NGSS 4-PS3 Energy (introductory) • NGSS K-2-ETS1

SDG badges



CLASS
25

My Talking Pet in Scratch

App, Web & Game Development • Software • Week 5

LE-25

Introduction

Virtual pets and talking toys respond when you play with them. Children choose a pet sprite in Scratch, record their own voice and program the pet to talk, jump or change costume when it is clicked or when a key is pressed. With the teacher's help they add a 'feed me' button. They learn that programs react to events and can be personal and funny.

Activities

GUIDED

- Record a sound and play it when the pet is clicked
- Make the pet jump and change costume with a key

DISCOVERY

- Add a hunger counter that goes up and down

Outcomes

Computing: Events, sounds and costumes

Arts: Character design and voice acting

Life Skills: Caring for a (virtual) pet



Software

Scratch 3 (teacher class account)

Also: ScratchJr

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1A) • ISTE Creative Communicator

SDG badges



Introduction

Every photo sent on a phone travels as numbers. Children colour squares on a grid to make a tiny picture, turn it into a row of numbers, and read the numbers to a partner who redraws the picture without seeing it. They discover pixels, codes and why one wrong number spoils the picture. It is their first look at how computers send pictures around the world.

Activities

GUIDED

- Turn an 8x8 picture into numbers
- Send it to a partner by reading the numbers aloud

DISCOVERY

- Make a colour code (1 = red, 2 = blue) and send a colour picture

Outcomes

Computing: Pictures are stored as numbers (pixels)

Maths: Grids, rows and columns

Technology: How phones send photos



Software

No software needed

Also: Code.org pixel lessons

Key hardware (per group)

- Picture cards and stationery

Standards

CSTA Data & Analysis (1A) • CSTA Networks & the Internet (1A)

SDG badges



CLASS
27

Pop the Bubbles with Your Hand

App, Web & Game Development + Sensors & Environment • Software • Week 6

LE-27

Introduction

Some games use a camera so you play by moving your body. Children build a Scratch game where bubbles float up and pop when they wave their hand in front of the webcam, using Scratch's video sensing. They count points and change the speed of the bubbles. They discover that the computer is 'watching' for movement, just like an automatic door.

Activities

GUIDED

- Make a bubble float across the stage
- Pop it with video motion and add a score

DISCOVERY

- Add a second bubble and a timer

Outcomes

Computing: Camera input and variables

Science: Movement can be detected

Life Skills: Healthy active play



Software

Scratch 3 (video sensing extension)

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1A) • ISTE Innovative Designer

SDG badges



Introduction

Before electronics, inventors built moving toys using cams: odd-shaped wheels that push things up and down. Children build a cardboard box automaton where turning a crank makes a bird, fish or robot bob up and down. They test round and egg-shaped cams and see how the shape changes the movement. They learn that machines turn one kind of motion into another.

Activities

GUIDED

- Assemble the box, axle and cam
- Turn the crank and watch the figure move

DISCOVERY

- Swap in a different cam shape and describe the new movement

Outcomes

Engineering: Cams change turning into up-and-down motion

Arts: Decorating a moving sculpture

Maths: Shapes and circles



Software

No software needed

Key hardware (per group)

- Picture cards and stationery

Standards

NGSS K-2-ETS1 • ISTE Innovative Designer

SDG badges



CLASS
29

Teach the Computer Our Sounds

AI & Data • Software • Week 7

LE-29

Introduction

Smart speakers learn to recognise voices and sounds. With the teacher, children record examples of claps, whistles and silence in Teachable Machine and train a sound model. They test it with new sounds and find out what confuses it, then record better examples. They learn that computers learn from examples, and more examples help.

Activities

GUIDED

- Record examples of three sounds
- Train the model and test it with new sounds

DISCOVERY

- Find a sound that tricks the model and add better examples

Outcomes

AI: Computers learn from examples

Science: Different sounds have different patterns

Life Skills: Taking turns and listening



Software

Teachable Machine (audio, teacher-led)

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Understand) • CSTA AI concepts

SDG badges



Introduction

Robot pets and luggage-carrying robots can follow a person. Children use a robot car with a distance sensor: the teacher uploads a starter program and children change the numbers so the 'pet' follows their hand, stops when too close and backs away. They test speeds and distances and give their pet a name and a face. They see how a sensor and a motor work together to make a robot seem alive.

Activities

GUIDED

- Make the robot follow a hand held in front of it
- Change the stop distance and speed numbers

DISCOVERY

- Decorate the robot as a pet and make it beep when happy

Outcomes

Engineering: Sensors guide motors

Computing: Changing numbers changes behaviour

Maths: Near and far, measuring in cm



Software

mBlock 5 (starter program, teacher-uploaded)

Key hardware (per group)

- 4WD robot car (assembled, L298N) x1
- Ultrasonic sensor HC-SR04 x1

Standards

NGSS K-2-ETS1 • CSTA Computing Systems (1A)

SDG badges



CLASS
31

Safety Helper Robot Design

Security & Safety + Mechanical Design • Hands-on (unplugged) • Week 8

LE-31

Introduction

Engineers design robots to keep people safe, like robots that warn of fire or help older people. Children choose a safety problem at home or school (a hot stove, a wet floor, a busy road) and design a helper robot on paper, deciding which sensors it needs and how it warns people. They build a cardboard model and present it. They think about safety, kindness and who their robot helps.



Software

No software needed

Key hardware (per group)

- Picture cards and stationery

Standards

NGSS K-2-ETS1 • ISTE Empowered Learner

SDG badges



Activities

GUIDED

- Choose a safety problem and draw a helper robot
- Label its sensors and how it warns people

DISCOVERY

- Build a cardboard model and present it to the class

Outcomes

Life Skills: Staying safe at home and school

Engineering: Matching sensors to problems

Arts: Designing and building a model

CLASS
32

Countdown Rocket Launch

App, Web & Game Development + Programming & Logic • Hardware + Software • Week 8

LE-32

Introduction

Space agencies count down before every launch. Children build a Scratch-style rocket animation in mBlock that counts down from 10 and blasts off when a real button on the Arduino is pressed, with LEDs flashing for the engines. They add sound effects and a launch message. Real buttons controlling an on-screen story make coding feel magical.



Software

mBlock 5 (live mode with sprites)

Also: Scratch 3

Key hardware (per group)

- Arduino Uno + USB cable x1
- Push button x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2
- Resistor 10k ohm x1

Standards

CSTA Algorithms & Programming (1A) • NGSS 1-ESS1 Space

SDG badges



Activities

GUIDED

- Make the rocket count down and fly on screen
- Launch it with the real Arduino button and flash the LEDs

DISCOVERY

- Add planets to visit and a safe landing

Outcomes

Computing: Counting variables and hardware events

Science: Rockets and space

Arts: Animation and sound effects



Phase 2 roadmap: classes 17-32

8 weeks, 2 classes a week: four phases of 4 classes. Builds on Phase 1; the capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Code Studio

Python Turtle art, a MakeCode Arcade game, binary secret codes and a joystick maze

C17



Python Turtle Art: Shapes and Patterns

Programming & Logic · **Software**

Python loops and angles

JM-17

C18



MakeCode Arcade: My First Platform Game

App, Web & Game Development · **Software**

Sprites, events, variables and gravity

JM-18

C19



Binary Bracelets & Secret Codes

IoT & Communication · **Hands-on (unplugged)**

Binary and encoding

JM-19

C20



Joystick Maze Game

App, Web & Game Development · **Hardware + Software**

Analog values and coordinates

JM-20

Phase 2

Weeks 3-4

Light, Sound & Music Tech

NeoPixel light shows, an LED matrix, an air theremin and Processing art

C21



NeoPixel Light Show to the Beat

Electronics & Circuits · **Hardware**

Loops and arrays of LEDs

JM-21

C22



8x8 LED Matrix Emoji Display

Electronics & Circuits · **Hardware**

Binary patterns and animation frames

JM-22

C23



Ultrasonic Theremin: Music in the Air

Sensors & Environment · **Hardware**

Pitch and sound waves

JM-23

C24



Processing: Interactive Drawing Machine

App, Web & Game Development · **Software**

setup(), draw(), coordinates and events

JM-24

Phase 3

Weeks 5-6

Smart City & Mechanisms

A smart traffic junction, a kitchen scale, a claw machine and wind turbines

C25



Smart Traffic Junction with Car Counter

Sensors & Environment · **Hardware**

Sensor-based control

JM-25

C26



Smart Kitchen Scale

Sensors & Environment · **Hardware**

Force, weight and mass

JM-26

C27



Arcade Claw Machine

Robotics & Actuation · **Hardware + Software**

Servos, linkages and grippers

JM-27

C28



Wind Turbine Power Lab

Power & Energy · **Hardware**

Energy transfer from wind to electricity

JM-28

Phase 4

Weeks 7-8

Code, Data & AI

A Python guessing game, class data detectives, an AI emotion mirror and a cyber escape room

C29



Python Number Guessing Game

Programming & Logic · **Software**

Variables, loops and conditions in Python

JM-29

C30



Data Detectives: Class Survey Charts

AI & Data · **Software**

Collecting, graphing and interpreting data

JM-30

C31



AI Emotion Mirror

AI & Data · **Software**

Training data, testing and bias

JM-31

C32



Cyber Safety Escape Room

Security & Safety · **Hands-on (unplugged)**

Passwords, phishing and privacy

JM-32



After class 32 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: A smart lunchbox scale that tracks food waste; A Python or MakeCode game for younger students; A light-show costume or poster; A smart crossing for the school gate. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
17**Python Turtle Art: Shapes and Patterns**

Programming & Logic • Software • Week 1

JM-17

Introduction

Artists and engineers use code to draw precise patterns. Students write their first Python program: a turtle that draws squares, stars and spirals with for loops. They change angles and colours to discover how a few lines of code make amazing patterns. They finish with a personal 'code art' poster.

**Software****Python Turtle (Thonny)**

Also: trinket.io

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1B) • ISTE Computational Thinker

SDG badges

Activities

GUIDED

- Draw a square and a star with forward and right
- Use a for loop and colours to draw a spiral

DISCOVERY

- Design a pattern of your own and explain the code

Outcomes

Computing: Python loops and angles**Maths:** Angles in shapes (360 degrees)**Arts:** Geometric artCLASS
18**MakeCode Arcade: My First Platform Game**

App, Web & Game Development • Software • Week 1

JM-18

Introduction

Many famous games are platform games where a character jumps between blocks. Students build their own in Microsoft MakeCode Arcade: a player sprite, gravity, jumping, coins to collect and an enemy to avoid. They draw their own sprite and level map. Classmates play-test each other's games.

**Software****MakeCode Arcade**

Also: Scratch 3

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1B) • ISTE Creative Communicator

SDG badges

Activities

GUIDED

- Make a player that moves and jumps on a tile map
- Add coins, a score and a game-over rule

DISCOVERY

- Design a second level that is harder

Outcomes

Computing: Sprites, events, variables and gravity**Arts:** Pixel art and level design**Life Skills:** Giving and using feedback

CLASS
19

Binary Bracelets & Secret Codes

IoT & Communication + Security & Safety · Hands-on (unplugged) · Week 2

JM-19

Introduction

Computers store every letter as 0s and 1s. Students turn their initials into binary using two colours of beads or paper strips, then send and crack secret messages with a Caesar cipher wheel. They learn how computers count in binary and how codes keep messages private. Everyone leaves with a wearable binary bracelet.



Software

No software needed

Also: CS Unplugged

Key hardware (per group)

- Picture cards and stationery

Standards

CSTA Data & Analysis (1B) · CSTA Networks & the Internet (1B)

SDG badges



Activities

GUIDED

- Convert your initials to binary and make a bracelet
- Encode and decode a Caesar cipher message

DISCOVERY

- Break a partner's code without the key

Outcomes

Computing: Binary and encoding**Maths:** Powers of 2 and place value**Life Skills:** Why privacy mattersCLASS
20

Joystick Maze Game

App, Web & Game Development + Electronics & Circuits · Hardware + Software · Week 2

JM-20

Introduction

Game controllers turn your hand movements into numbers. Students wire a joystick to an Arduino and use mBlock's live mode to steer a sprite through a maze they draw in Scratch style. They discover that the joystick sends X and Y numbers from 0 to 1023. Touching a wall sends the player back to the start.



Software

mBlock 5 (live mode)

Also: Scratch 3

Key hardware (per group)

- Arduino Uno + USB cable x1
- Joystick module x1

Standards

CSTA Computing Systems (1B) · ISTE Innovative Designer

SDG badges



Activities

GUIDED

- Wire the joystick and read X and Y values
- Move the sprite through a maze using the joystick

DISCOVERY

- Add a timer and a high score

Outcomes

Computing: Analog values and coordinates**Engineering:** Building a game controller**Maths:** X and Y coordinates

CLASS
21

NeoPixel Light Show to the Beat

Electronics & Circuits + Sensors & Environment • Hardware • Week 3

JM-21

Introduction

Concert lights flash in time with the music. Students program a strip of 30 colour LEDs to make rainbows, chasers and fire effects, then add a sound sensor so the lights pulse with claps and music. They learn how one data wire controls many LEDs. Each team performs a 30-second light show.

Activities

GUIDED

- Program rainbow and chaser effects on the LED strip
- Make the strip react to sound from the sound sensor

DISCOVERY

- Choreograph a light show for a song

Outcomes

Computing: Loops and arrays of LEDs**Science:** Sound levels and colour**Arts:** Light design**Software****ArduinoBlocks**

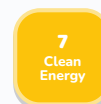
Also: Wokwi (NeoPixel)

Key hardware (per group)

- Arduino Uno + USB cable x1
- WS2812B LED strip (1 m, 30 LEDs) x1
- Sound sensor KY-037 x1
- Capacitor 100 uF x1
- Resistor 220 ohm x1

Standards

CSTA Algorithms & Programming (1B) • NGSS 4-PS4 Waves

SDG badgesCLASS
22

8x8 LED Matrix Emoji Display

Electronics & Circuits + Programming & Logic • Hardware • Week 3

JM-22

Introduction

Scoreboards and bus signs are made of grids of LEDs. Students draw emojis and letters on squared paper, convert each row into binary and send them to an 8x8 LED matrix. They animate a winking face and scroll their name across the display. Binary from the bracelet lesson suddenly becomes useful.

Activities

GUIDED

- Design an emoji on a grid and convert it to 8 binary rows
- Display and animate it on the matrix

DISCOVERY

- Scroll a message across the display

Outcomes

Computing: Binary patterns and animation frames**Maths:** Grids and binary numbers**Arts:** Pixel design**Software****ArduinoBlocks**

Also: Wokwi (MAX7219)

Key hardware (per group)

- Arduino Uno + USB cable x1
- MAX7219 8x8 LED matrix x1

Standards

CSTA Data & Analysis (1B) • CSTA Algorithms & Programming (1B)

SDG badges

CLASS
23

Ultrasonic Theremin: Music in the Air

Sensors & Environment • Hardware • Week 4

JM-23

Introduction

The theremin is a musical instrument you play without touching it. Students build one with an ultrasonic sensor and a passive buzzer: moving a hand closer or further changes the note. They map distance to pitch and add buttons for different 'instruments'. They perform a simple tune in the air.



Activities

GUIDED

- Read hand distance with the ultrasonic sensor
- Map the distance to a musical note

DISCOVERY

- Snap notes to a real scale so tunes sound better

Outcomes

Science: Pitch and sound waves**Computing:** Mapping one range to another**Arts:** Performing music**Software****ArduinoBlocks**

Also: Tinkercad Circuits

Key hardware (per group)

- Arduino Uno + USB cable x1
- Ultrasonic sensor HC-SR04 x1
- Passive buzzer x1
- Push button x1
- Resistor 10k ohm x1

Standards

NGSS 4-PS4 Waves • CSTA Algorithms & Programming (1B)

SDG badgesCLASS
24

Processing: Interactive Drawing Machine

App, Web & Game Development • Software • Week 4

JM-24

Introduction

Processing is a free coding language made for artists and designers. Students write their first Processing sketch: a canvas where the mouse paints with shapes, colours change with the mouse position and a key clears the screen. They learn the setup and draw functions that run every frame. Each student exports a digital artwork for the class gallery.



Activities

GUIDED

- Draw shapes that follow the mouse in Processing
- Change colour with mouse position and clear with a key

DISCOVERY

- Add a rainbow brush or a symmetry mirror

Outcomes

Computing: setup(), draw(), coordinates and events**Arts:** Digital art**Maths:** X and Y coordinates on a canvas**Software****Processing 4**

Also: p5.js web editor

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1B) • ISTE Creative Communicator

SDG badges

CLASS
25

Smart Traffic Junction with Car Counter

Sensors & Environment + Programming & Logic • Hardware • Week 5

JM-25

Introduction

Smart cities change traffic lights depending on how many cars are waiting. Students build a model junction with two sets of traffic lights and IR sensors that count toy cars on each road. The road with more cars gets a longer green light, and the counts appear on an OLED. They compare smart timing with fixed timing.

Activities

GUIDED

- Count cars with IR sensors and show counts on the OLED
- Give the busier road a longer green light

DISCOVERY

- Add a pedestrian button that safely stops both roads

Outcomes

Engineering: Sensor-based control**Computing:** Counting and comparing variables**Life Skills:** Road safety

Software

ArduinoBlocks

Also: Tinkercad Circuits

Key hardware (per group)

- Arduino Uno + USB cable x1
- IR sensor module x2
- LED 5 mm (assorted) x6
- Resistor 220 ohm x6
- OLED 0.96 in I2C x1

Standards

NGSS 3-5-ETS1 • CSTA Algorithms & Programming (1B)

SDG badges



Introduction

Bakers, shops and post offices weigh things precisely. Students build a digital scale with a load cell and an amplifier board, calibrate it with known weights (a full water bottle, coins) and show grams on an OLED. They add a 'tare' button and track how much food is wasted at lunch. They learn that sensors must be calibrated to be trusted.

Activities

GUIDED

- Wire the load cell and calibrate it with a known weight
- Show the weight in grams and add a tare button

DISCOVERY

- Weigh lunch leftovers and present a food-waste chart

Outcomes

Science: Force, weight and mass**Maths:** Calibration and units**Life Skills:** Reducing food waste

Software

ArduinoBlocks

Also: Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- Load cell 5 kg + HX711 x1
- OLED 0.96 in I2C x1
- Push button x1
- Resistor 10k ohm x1

Standards

NGSS 3-5-ETS1 • CSTA Data & Analysis (1B)

SDG badges



CLASS
27

Arcade Claw Machine

Robotics & Actuation + Mechanical Design • Hardware + Software • Week 6

JM-27

Introduction

Claw machines are robots controlled by a joystick. Students build a cardboard claw machine: one servo swings the arm, another opens and closes a claw made from ice-cream sticks, and a joystick controls them both. They tune the claw so it can pick up a small toy and add a 30-second timer. Then they challenge each other to win a prize.

Activities

GUIDED

- Build the claw and arm with two servos
- Control both servos with the joystick

DISCOVERY

- Add a timer and a 'drop' button

Outcomes

Engineering: Servos, linkages and grippers**Computing:** Mapping joystick values to angles**Life Skills:** Fair play and game design**Software****ArduinoBlocks**

Also: Tinkercad Circuits

Key hardware (per group)

- Arduino Uno + USB cable x1
- Servo SG90 x2
- Joystick module x1

Standards

NGSS 3-5-ETS1 • CSTA Algorithms & Programming (1B)

SDG badges

Introduction

Wind farms turn moving air into electricity. Students design turbine blades from card, fix them to a motor that acts as a generator and measure the voltage with a multimeter in front of a fan. They run a fair test changing only the number or angle of the blades. They graph their results and find the best design.

Activities

GUIDED

- Build blades and attach them to the generator motor
- Measure voltage for different blade designs

DISCOVERY

- Graph the results and explain the best design

Outcomes

Science: Energy transfer from wind to electricity**Engineering:** Fair testing and design iteration**Maths:** Recording and graphing data**Software****No software needed**

Also: PhET Energy Forms

Key hardware (per group)

- DC hobby motor x1
- 5 V DC fan x1 (class)

Standards

NGSS 4-PS3 Energy • NGSS 3-5-ETS1

SDG badges

CLASS
29

Python Number Guessing Game

Programming & Logic • Software • Week 7

JM-29

Introduction

Every game needs logic: was the guess too high, too low or just right? Students write a Python game where the computer picks a secret number and gives hints until the player guesses it. They use variables, while loops, if-elif-else and random numbers, and then count the guesses. They discover the 'halving' strategy that always wins quickly.

Activities

GUIDED

- Pick a random number and check a guess
- Loop with hints until the player wins

DISCOVERY

- Count guesses and add an easy and a hard mode

Outcomes

Computing: Variables, loops and conditions in Python**Maths:** Halving strategy (binary search)**Life Skills:** Logical thinking**Software****Python (Thonny)**

Also: trinket.io

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (1B) • ISTE Computational Thinker

SDG badges

Introduction

Scientists, doctors and businesses make decisions from data. Students design a short class survey (favourite fruit, minutes of screen time, travel to school), collect answers, and explore them in CODAP with tables and graphs. They spot patterns, find an outlier and present one surprising fact. They learn that good questions make good data.

Activities

GUIDED

- Design a survey and collect answers from the class
- Make bar charts and dot plots in CODAP

DISCOVERY

- Find a pattern and present it as a news headline

Outcomes

Maths: Collecting, graphing and interpreting data**Computing:** Data tools**Life Skills:** Asking fair questions**Software****CODAP**

Also: Google Sheets

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Data & Analysis (1B) • ISTE Knowledge Constructor

SDG badges

CLASS
31**AI Emotion Mirror**

AI & Data • Software • Week 8

JM-31

Introduction

Some apps can recognise faces and expressions. Students train an image model to tell 'happy', 'surprised' and 'thinking' faces apart, then connect it to a Scratch-style program in PictoBlox so a character copies their expression. They test it on different classmates and discuss when face-reading AI could be unfair or invasive.

Activities

GUIDED

- Collect face examples and train the model
- Make a sprite react to each expression

DISCOVERY

- Test with many people and record when it fails

Outcomes

AI: Training data, testing and bias**Computing:** Connecting a model to a program**Life Skills:** Privacy and consent with faces**Software****PictoBlox (machine learning extension)**

Also: Teachable Machine

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Understand/Apply) • CSTA AI concepts

SDG badges

Introduction

Online criminals try to trick people into sharing passwords and personal information. Teams race through an escape-room of puzzles: spot the fake message, build a strong password, crack a code lock and decide what is safe to share online. Each solved puzzle reveals a clue to the final code. They leave with practical rules to stay safe online.

Activities

GUIDED

- Solve phishing, password and privacy puzzles
- Combine the clues to open the final lock

DISCOVERY

- Design one new puzzle for next year's class

Outcomes

Computing: Passwords, phishing and privacy**Life Skills:** Staying safe online**Maths:** Codes and logic puzzles**Software****No software needed**

Also: Interland (Google Be Internet Awesome)

Key hardware (per group)

- 4x4 matrix keypad x1 (class)

Standards

CSTA Impacts of Computing (1B) • ISTE Digital Citizen

SDG badges



Phase 2 roadmap: classes 17-32

8 weeks, 2 classes a week: four phases of 4 classes. Builds on Phase 1; the capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Creative Coding

Processing art and radar, a Pygame game and an ESP32-CAM wildlife camera

C17



Processing: Generative Art Poster

App, Web & Game Development • **Software**

Nested loops, random numbers and functions

YI-17

C18



Processing + Arduino: Live Sensor Radar

Programming & Logic • **Hardware + Software**

Serial communication and parsing

YI-18

C19



Python Pygame: Catch the Falling Stars

App, Web & Game Development • **Software**

Game loops, collisions, lists

YI-19

C20



ESP32-CAM Wildlife Camera

IoT & Communication • **Hardware**

Wi-Fi streaming and triggers

YI-20

Phase 2

Weeks 3-4

Data & Signals

Python data logging, a posture wearable, Li-Fi messages and code breaking

C21



Python Data Lab: Is Our Classroom Too Hot?

AI & Data • **Software**

Files, lists and plotting in Python

YI-21

C22



Smart Posture Corrector Wearable

Sensors & Environment • **Hardware**

Posture and health

YI-22

C23



Li-Fi Messenger: Sending Text with Light

IoT & Communication • **Hardware**

Binary encoding and timing

YI-23

C24



Python Code Breakers: Caesar to Vigenère

Security & Safety • **Software**

Strings, loops, functions and dictionaries

YI-24

Phase 3

Weeks 5-6

Robots & Mechanisms

Robot sumo, a walking linkage robot, an energy meter and a Python control panel

C25



Robot Sumo Challenge

Robotics & Actuation • **Hardware**

Sensors, motors and strategy

YI-25

C26



Walking Linkage Robot

Mechanical Design • **Hardware**

Linkages and cranks

YI-26

C27



Energy Detective: How Much Power Do Gadgets Use?

Power & Energy • **Hardware**

Voltage, current, power and energy

YI-27

C28



Python Control Panel for Arduino

Programming & Logic • **Hardware + Software**

GUIs, events and serial communication

YI-28

Phase 4

Weeks 7-8

AI & the Web

A voice-controlled p5.js game, a weather API dashboard, sign-language AI and a custom controller

C29



p5.js: Flappy Voice Game

App, Web & Game Development • **Software**

JavaScript, objects and the game loop

YI-29

C30



Python Weather API Dashboard

IoT & Communication • **Software**

APIs, JSON and dictionaries

YI-30

C31



Sign Language Alphabet Recogniser

AI & Data • **Software**

Training, testing and accuracy

YI-31

C32



Build Your Own Game Controller

App, Web & Game Development • **Hardware + Software**

Human-centred controller design

YI-32



After class 32 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: A data-logging study room monitor with charts; An accessible game controller for a classmate; A wildlife camera survey of the school garden; An AI tool that helps someone communicate. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
17**Processing: Generative Art Poster**

App, Web & Game Development • Software • Week 1

YI-17

Introduction

Generative artists write code that creates artwork they could never draw by hand. Students use Processing to build posters from loops, randomness, noise and colour palettes: grids of shapes, flowing lines and particle trails. Every run makes a different design. They export a high-resolution poster and explain the rules behind it.

**Software****Processing 4**

Also: p5.js web editor

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (2) • ISTE Creative Communicator

SDG badges

Activities

GUIDED

- Draw a grid of shapes with nested loops
- Use random() and noise() to make each run unique

DISCOVERY

- Print the best design as an A4 poster

Outcomes

Computing: Nested loops, random numbers and functions**Arts:** Generative design**Maths:** Coordinates and transformationsCLASS
18**Processing + Arduino: Live Sensor Radar**

Programming & Logic + Sensors & Environment • Hardware + Software • Week 1

YI-18

Introduction

Radar screens draw what sensors detect in real time. Students mount an ultrasonic sensor on a servo that sweeps 180 degrees, send angle and distance over serial, and write a Processing sketch that draws a green radar screen with detected objects. They learn how two programs on two devices exchange data.

**Software**

Arduino IDE 2 + Processing 4

Key hardware (per group)

- Arduino Uno + USB cable x1
- Ultrasonic sensor HC-SR04 x1
- Servo SG90 x1

Standards

CSTA Computing Systems (2) • CSTA Algorithms & Programming (2)

SDG badges

Activities

GUIDED

- Sweep the servo and send angle,distance over serial
- Draw the radar sweep and objects in Processing

DISCOVERY

- Colour objects by distance and add a 'too close' alarm

Outcomes

Computing: Serial communication and parsing**Maths:** Angles, sine and cosine**Engineering:** Sensor scanning systems

CLASS
19**Python Pygame: Catch the Falling Stars**

App, Web & Game Development • Software • Week 2

YI-19

Introduction

Python is used to build real games. Students write a Pygame game where a basket catches falling stars: a game loop, keyboard control, collision detection, score and lives. They add sound effects and increasing difficulty. Classmates play-test and suggest improvements.

**Software**

Python + Pygame (Thonny)

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (2) • ISTE Computational Thinker

SDG badges

Activities

GUIDED

- Build the game loop with a moving player
- Add falling objects, collisions and score

DISCOVERY

- Add levels, sounds and a high score

Outcomes

Computing: Game loops, collisions, lists

Maths: Coordinates and speed

Life Skills: Iterating from feedback

CLASS
20**ESP32-CAM Wildlife Camera**

IoT & Communication + Sensors & Environment • Hardware • Week 2

YI-20

Introduction

Conservationists use hidden 'camera traps' to study wild animals. Students set up an ESP32-CAM that streams video to a browser over Wi-Fi, then add a PIR sensor so it snaps a photo when something moves. They test placement around the school garden and discuss privacy rules for cameras.

**Software**

Arduino IDE 2 (ESP32-CAM)

Key hardware (per group)

- ESP32-CAM + programmer x1
- PIR motion sensor x1

Standards

CSTA Networks & the Internet (2) • NGSS MS-LS2 Ecosystems

SDG badges

Activities

GUIDED

- Program the ESP32-CAM and view the live stream
- Trigger a photo with the PIR sensor

DISCOVERY

- Place it outdoors and review what it captured

Outcomes

Computing: Wi-Fi streaming and triggers

Science: Observing wildlife

Life Skills: Camera privacy and consent

CLASS
21**Python Data Lab: Is Our Classroom Too Hot?**

AI & Data • Software • Week 3

YI-21

Introduction

Scientists log data over time to answer real questions. Students record temperature and humidity from a DHT22 to a CSV file every minute during a school day, then analyse it in Python with matplotlib: averages, peaks and graphs. They compare rooms and recommend when to open windows or switch on fans.

**Software****Python (Thonny + matplotlib)**

Also: Google Sheets

Key hardware (per group)

- Arduino Uno + USB cable x1
- DHT22 sensor x1

Standards

CSTA Data & Analysis (2) • NGSS MS-ETS1

SDG badges

Activities

GUIDED

- Log DHT22 readings to a CSV file
- Plot and analyse the data in Python

DISCOVERY

- Compare two rooms and write a recommendation

Outcomes

Computing: Files, lists and plotting in Python**Science:** Heat and comfort**Maths:** Mean, maximum and trendsCLASS
22**Smart Posture Corrector Wearable**

Sensors & Environment • Hardware • Week 3

YI-22

Introduction

Bad posture while studying can cause back pain. Students attach an MPU6050 motion sensor to a strap, read the tilt angle and make a vibration motor buzz when they slouch for too long. They calibrate a 'good posture' angle and test it while working. They discuss health data and wearables.

**Software****Arduino IDE 2**

Also: Wokwi (MPU6050)

Key hardware (per group)

- Arduino Uno + USB cable x1
- MPU6050 IMU x1
- Vibration motor module x1
- OLED 0.96 in I2C x1

Standards

NGSS MS-ETS1 • CSTA Computing Systems (2)

SDG badges

Activities

GUIDED

- Read tilt angle from the MPU6050
- Vibrate after slouching for 5 seconds

DISCOVERY

- Count slouches per hour and show them on the OLED

Outcomes

Science: Posture and health**Computing:** Sensor angles and timers**Engineering:** Wearable design

CLASS
23**Li-Fi Messenger: Sending Text with Light**

IoT & Communication + Electronics & Circuits • Hardware • Week 4

YI-23

Introduction

Li-Fi sends data with flickering light instead of radio. Students build a transmitter that blinks an LED in binary and a receiver on a second Arduino that reads it with an LDR and decodes the text. They test speed and distance and see how errors appear. They compare light, radio and cable communication.

**Software**

Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x2
- LED 5 mm (assorted) x1
- LDR module x1
- Resistor 220 ohm x1

Standards

CSTA Networks & the Internet (2) • NGSS MS-PS4 Waves

SDG badges**Activities****GUIDED**

- Send letters as binary light pulses
- Receive and decode them with an LDR

DISCOVERY

- Add a checksum to detect errors

Outcomes**Computing:** Binary encoding and timing**Science:** Light as a signal**Engineering:** Two-device communicationCLASS
24**Python Code Breakers: Caesar to Vigenère**

Security & Safety • Software • Week 4

YI-24

Introduction

Codes have protected secrets for thousands of years. Students write Python programs to encrypt and decrypt Caesar and Vigenère ciphers, then break a Caesar message by brute force and frequency analysis. They learn why modern encryption needs computers and long keys.

**Software**

Python (Thonny)

Also: CryptoClub

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Networks & the Internet (2) • CSTA Algorithms & Programming (2)

SDG badges**Activities****GUIDED**

- Write encrypt and decrypt functions for a Caesar cipher
- Break a secret message with frequency analysis

DISCOVERY

- Implement Vigenère and explain why it is harder to break

Outcomes**Computing:** Strings, loops, functions and dictionaries**Maths:** Modular arithmetic**Life Skills:** Why encryption protects us

CLASS
25

Robot Sumo Challenge

Robotics & Actuation + Sensors & Environment • Hardware • Week 5

YI-25

Introduction

Robot sumo is a real competition: robots try to push each other out of a ring. Teams program a 4WD car to find the opponent with an ultrasonic sensor and stay inside the ring using IR edge sensors. They design a strategy, tune speeds and compete in a class tournament.

Activities

GUIDED

- Detect the ring edge with IR sensors and reverse
- Find and push the opponent with the ultrasonic sensor

DISCOVERY

- Design a winning strategy and add a wedge

Outcomes

Engineering: Sensors, motors and strategy**Computing:** State machines**Life Skills:** Fair competition**Software**

Arduino IDE 2

Key hardware (per group)

- 4WD robot car (assembled, L298N) x1
- Ultrasonic sensor HC-SR04 x1
- IR sensor module x2

Standards

NGSS MS-ETS1 • CSTA Algorithms & Programming (2)

SDG badgesCLASS
26

Walking Linkage Robot

Mechanical Design • Hardware • Week 5

YI-26

Introduction

Engineers copy nature: some robots walk instead of rolling. Students build a crank-driven walking mechanism from ice-cream sticks or 3D-printed links, inspired by Theo Jansen's Strandbeest, and drive it with a geared motor. They measure stride length and improve the linkage. They discover how one turning motor creates a walking motion.

Activities

GUIDED

- Build one leg linkage and test its path
- Join legs and drive them with a motor

DISCOVERY

- Change link lengths and measure the new stride

Outcomes

Engineering: Linkages and cranks**Maths:** Lengths and motion paths**Arts:** Kinetic sculpture**Software****No software needed**

Also: Tinkercad 3D (printed links)

Key hardware (per group)

- DC hobby motor x1
- 4xAA battery pack x1

Standards

NGSS MS-ETS1 • ISTE Innovative Designer

SDG badges

CLASS
27**Energy Detective: How Much Power Do Gadgets Use?**

Power & Energy + Electronics & Circuits • Hardware • Week 6

YI-27

Introduction

Electricity bills depend on how much power our devices use. Students use an INA219 current sensor to measure voltage, current and power of LEDs, a motor, a fan and a phone charging, then calculate energy and cost over a day. They rank gadgets and suggest savings for their homes.

**Software**

Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- Current sensor INA219 x1
- OLED 0.96 in I2C x1
- LED 5 mm (assorted) x2
- DC hobby motor x1
- 4xAA battery pack x1

Standards

NGSS MS-PS3 Energy • CSTA Data & Analysis (2)

SDG badges

Activities

GUIDED

- Measure current and power with the INA219
- Calculate energy used in a day

DISCOVERY

- Estimate the monthly cost and suggest savings

Outcomes

Science: Voltage, current, power and energy**Maths:** Rates, units and cost**Life Skills:** Saving energy at homeCLASS
28**Python Control Panel for Arduino**

Programming & Logic + App, Web & Game Development • Hardware + Software • Week 6

YI-28

Introduction

Real machines have control panels on a computer screen. Students write a Python program with a Tkinter window of buttons and sliders that sends commands over serial to an Arduino, turning LEDs on and moving a servo, and shows a live sensor reading. They learn about GUIs and how software talks to hardware.

**Software**

Python (Thonny + pyserial, tkinter) + Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2
- Servo SG90 x1
- LDR module x1

Standards

CSTA Computing Systems (2) • CSTA Algorithms & Programming (2)

SDG badges

Activities

GUIDED

- Build a Tkinter window with buttons and a slider
- Send commands to the Arduino with pyserial

DISCOVERY

- Show a live sensor value in the window

Outcomes

Computing: GUIs, events and serial communication**Engineering:** Human-machine interfaces**Arts:** Interface layout

CLASS
29**p5.js: Flappy Voice Game**

App, Web & Game Development • Software • Week 7

YI-29

Introduction

Some games are controlled by your voice. Students code a Flappy-style game in p5.js (JavaScript) where the bird flies up when you make a sound into the microphone. They learn JavaScript basics, game objects and reading the microphone level, then share a link that runs in any browser.

Activities

GUIDED

- Make the bird fall and fly with p5.js
- Read the microphone level to control the bird

DISCOVERY

- Add pipes, collisions and a score

Outcomes

Computing: JavaScript, objects and the game loop

Science: Sound volume

Arts: Game feel and design

**Software**

p5.js web editor

Also: Processing 4

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (2) • ISTE Creative Communicator

SDG badges

Introduction

Weather apps fetch data from online services called APIs. Students write Python code that requests a 7-day forecast for Islamabad from the free Open-Meteo API, reads the JSON reply, and plots temperature and rain with matplotlib. They compare the forecast with their own sensor readings.

Activities

GUIDED

- Request the forecast and read the JSON data
- Plot temperature and rain with matplotlib

DISCOVERY

- Compare the forecast with a DHT22 reading at school

Outcomes

Computing: APIs, JSON and dictionaries

Science: Weather forecasting

Maths: Reading graphs

**Software**

Python (Thonny + requests, matplotlib)

Also: Google Colab

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Networks & the Internet (2) • CSTA Data & Analysis (2)

SDG badges

CLASS
31

Sign Language Alphabet Recogniser

AI & Data + App, Web & Game Development • Software • Week 8

YI-31

Introduction

AI can help people communicate. Students train an image model to recognise a few sign-language hand shapes, export it, and build a small web page that shows the matching letter. They test it with different hands and lighting, measure accuracy and discuss why including deaf users in the design matters.

Activities

GUIDED

- Collect hand-shape images and train the model
- Export it and show letters on a web page

DISCOVERY

- Measure accuracy on new people and improve the data

Outcomes

AI: Training, testing and accuracy

Life Skills: Inclusive design

Computing: Using a model in a web page



Software

Teachable Machine + p5.js web editor

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Apply) • CSTA AI concepts

SDG badges



Introduction

Game companies design custom controllers for special games. Students build a controller from a joystick, buttons and a motion sensor on an Arduino, send the values over serial and use them to play their own Pygame game from YI-19. They test comfort and accessibility with classmates.

Activities

GUIDED

- Read joystick, buttons and tilt and send them over serial
- Control the Pygame game with the new controller

DISCOVERY

- Design a one-handed or accessible version

Outcomes

Engineering: Human-centred controller design

Computing: Serial data and input handling

Life Skills: Designing for accessibility



Software

Arduino IDE 2 + Python Pygame

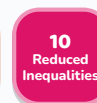
Key hardware (per group)

- Arduino Uno + USB cable x1
- Joystick module x1
- Push button x2
- Resistor 10k ohm x2
- MPU6050 IMU x1

Standards

CSTA Computing Systems (2) • ISTE Innovative Designer

SDG badges





Phase 2 roadmap: classes 17-32

8 weeks, 2 classes a week: four phases of 4 classes. Builds on Phase 1; the capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Vision & Intelligence

OpenCV tracking, a vision-guided car, TinyML gestures and flocking simulations

C17



Computer Vision with OpenCV: Colour Tracking

AI & Data · **Software**

Computer vision basics

TL-17

C18



Vision-Guided Robot Car

Robotics & Actuation · **Hardware + Software**

Closed-loop control with vision

TL-18

C19



TinyML Gesture Recogniser on ESP32

AI & Data · **Hardware + Software**

Datasets, features, training and deployment

TL-19

C20



Processing: Flocking Simulation (Boids)

Programming & Logic · **Software**

Objects, vectors and simulation

TL-20

Phase 2

Weeks 3-4

Algorithms & Systems

Pathfinding algorithms, Git teamwork, a solar microgrid and noise analysis with FFT

C21



Pathfinding Visualiser: BFS and A*

Programming & Logic · **Software**

Graphs, queues and search algorithms

TL-21

C22



Git & GitHub for Engineering Teams

Programming & Logic · **Software**

Version control and collaboration

TL-22

C23



Smart Microgrid Model

Power & Energy · **Hardware**

Energy systems and load priority

TL-23

C24



Noise Pollution Analyser with FFT

Sensors & Environment · **Software**

Sound frequency and health

TL-24

Phase 3

Weeks 5-6

Robots & Control

Arm inverse kinematics, a robot swarm, a 3D-printed gearbox and a smart cane

C25



Robotic Arm Inverse Kinematics

Robotics & Actuation · **Hardware + Software**

Trigonometry and the law of cosines

TL-25

C26



Robot Swarm: Leader and Follower Cars

Robotics & Actuation · **Hardware + Software**

Multi-robot coordination

TL-26

C27



3D-Printed Gearbox and Gear Ratios

Mechanical Design · **Hardware**

Gear ratios, torque and speed

TL-27

C28



Smart Cane for the Visually Impaired

Electronics & Circuits · **Hardware**

Assistive device design

TL-28

Phase 4

Weeks 7-8

AI, Data & Product

Pakistan air-quality data science, a neural network from scratch, Godot games and password security

C29



Air Quality Data Science for Pakistan

AI & Data · **Software**

Data cleaning and exploratory analysis

TL-29

C30



Neural Network from Scratch

AI & Data · **Software**

Neurons, weights, loss and training

TL-30

C31



2D Game Development in Godot

App, Web & Game Development · **Software**

Scenes, nodes, signals and scripting

TL-31

C32



Password Security Lab: Hashing and Attacks

Security & Safety · **Software**

Hashing, salts and brute force

TL-32



After class 32 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: A vision-guided sorting robot; A smog awareness dashboard for our city; An assistive device tested with a real user; A complete Godot game released to the school. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
17**Computer Vision with OpenCV: Colour Tracking**

AI & Data • Software • Week 1

TL-17

Introduction

Self-driving cars and factory robots see the world through cameras. Students use Python and OpenCV to read a webcam, convert frames to HSV colour space, track a coloured ball, draw its position and calculate its speed. They tune colour thresholds and handle lighting changes.

**Software**

Python + OpenCV

Also: Google Colab

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (3A) • UNESCO AI Competency (Apply)

SDG badges

Activities

GUIDED

- Capture webcam frames and isolate one colour
- Find the object's centre and draw its path

DISCOVERY

- Track two colours and count when they cross a line

Outcomes

AI: Computer vision basics**Computing:** Image arrays and thresholds**Maths:** Coordinates and speedCLASS
18**Vision-Guided Robot Car**

Robotics & Actuation + AI & Data • Hardware + Software • Week 1

TL-18

Introduction

Robots can steer by what they see. Students stream video from an ESP32-CAM mounted on a 4WD car to a laptop, process it with OpenCV to find a coloured target, and send steering commands back to the car over Wi-Fi. They measure delay and tune the controller so the car follows the target smoothly.

**Software**

Python + OpenCV + Arduino IDE 2 (ESP32)

Key hardware (per group)

- 4WD robot car (assembled, L298N) x1
- ESP32-CAM + programmer x1
- ESP32 Dev Board + USB cable x1

Standards

CSTA Networks & the Internet (3A) • NGSS HS-ETS1

SDG badges

Activities

GUIDED

- Stream ESP32-CAM video to OpenCV
- Send left/right/forward commands back over Wi-Fi

DISCOVERY

- Reduce lag and add a 'lost target' search routine

Outcomes

Engineering: Closed-loop control with vision**Computing:** Networking and real-time processing**AI:** Object tracking

CLASS
19**TinyML Gesture Recogniser on ESP32**

AI & Data + IoT & Communication • Hardware + Software • Week 2

TL-19

Introduction

TinyML runs machine learning on tiny chips without the internet. Students collect motion data from an MPU6050 for gestures like 'circle', 'shake' and 'idle', train a model in Edge Impulse, and deploy it to an ESP32 that recognises gestures live. They measure accuracy, model size and response time.

**Software**

Edge Impulse + Arduino IDE 2 (ESP32)

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- MPU6050 IMU x1

Standards

UNESCO AI Competency (Create) • CSTA AI concepts

SDG badges

Activities

GUIDED

- Collect and label gesture data from the MPU6050
- Train, deploy and test the model on the ESP32

DISCOVERY

- Use a gesture to control an LED or a servo

Outcomes

AI: Datasets, features, training and deployment**Computing:** Embedded constraints (memory, speed)**Engineering:** Wearable sensingCLASS
20**Processing: Flocking Simulation (Boids)**

Programming & Logic • Software • Week 2

TL-20

Introduction

Birds and fish move in beautiful flocks with no leader. Students code Craig Reynolds' Boids in Processing using vectors and three simple rules: separation, alignment and cohesion. They add obstacles and a predator and observe emergent behaviour. Swarm robots and drone light shows use the same ideas.

**Software****Processing 4**

Also: p5.js web editor

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (3B) • NGSS HS-LS2

SDG badges

Activities

GUIDED

- Code boids with position and velocity vectors
- Implement separation, alignment and cohesion

DISCOVERY

- Add a predator and obstacles and tune the weights

Outcomes

Computing: Objects, vectors and simulation**Maths:** Vector addition and normalisation**Science:** Emergent behaviour in nature

CLASS
21**Pathfinding Visualiser: BFS and A***

Programming & Logic • Software • Week 3

TL-21

Introduction

Map apps and robots find the shortest path around obstacles. Students build a Pygame grid where they draw walls, then animate breadth-first search and A* finding a route. They compare how many cells each algorithm explores and why heuristics help.

**Software**

Python + Pygame

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (3B) • ISTE Computational Thinker

SDG badges

Activities

GUIDED

- Build a clickable grid with start, goal and walls
- Animate BFS and A* and compare them

DISCOVERY

- Add diagonal moves or weighted terrain

Outcomes

Computing: Graphs, queues and search algorithms

Maths: Distance heuristics

Engineering: Robot navigation

CLASS
22**Git & GitHub for Engineering Teams**

Programming & Logic + App, Web & Game Development • Software • Week 3

TL-22

Introduction

Professional engineers never email code to each other: they use version control. Students learn Git (commit, branch, merge, pull request) by building a shared project page, resolving a merge conflict and reviewing each other's code. They set up a portfolio repository for all their club projects.

**Software**

Git + GitHub (13+ accounts)

Also: GitHub Desktop

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (3A) • ISTE Global Collaborator

SDG badges

Activities

GUIDED

- Commit, branch and merge in a team repository
- Open and review a pull request

DISCOVERY

- Resolve a merge conflict and publish a README

Outcomes

Computing: Version control and collaboration

Life Skills: Professional teamwork and code review

Engineering: Documentation

CLASS
23**Smart Microgrid Model**

Power & Energy + IoT & Communication • Hardware • Week 4

TL-23

Introduction

Microgrids share solar power between homes and decide which loads to run. Students build a model with a solar panel, 18650 battery and an ESP32 that measures battery voltage and switches 'house' loads on relays by priority when energy is low. They write an energy-management algorithm and test cloudy-day scenarios.

Activities

GUIDED

- Measure solar and battery voltage with the ESP32
- Switch loads by priority when the battery is low

DISCOVERY

- Log energy data and test a 'cloudy day' scenario

Outcomes

Engineering: Energy systems and load priority

Computing: Decision algorithms

Science: Solar energy and batteries

**Software**

Arduino IDE 2 (ESP32)

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- Small solar panel x1
- 18650 cell + holder x1
- TP4056 charger module x1
- Relay module (4-channel) x1
- Low-voltage LED lamp (5 V) x2
- Resistor 10k ohm x2

Standards

NGSS HS-ETS1 • NGSS HS-PS3

SDG badges

Introduction

Noise pollution affects sleep, learning and health. Students record sound with a laptop microphone, use Python and NumPy's Fast Fourier Transform to see which frequencies are present, and plot a live spectrum. They measure classroom, corridor and road noise and propose ways to reduce it.

Activities

GUIDED

- Record audio and plot the waveform
- Use an FFT to plot the frequency spectrum

DISCOVERY

- Compare locations and write a noise report

Outcomes

Science: Sound frequency and health

Computing: Arrays and signal processing

Maths: Frequency and the Fourier transform (intuition)

**Software**

Python (NumPy, matplotlib, sounddevice)

Also: Audacity

Key hardware (per group)

- Laptop (1 per group)

Standards

NGSS HS-PS4 Waves • CSTA Data & Analysis (3A)

SDG badges

CLASS
25**Robotic Arm Inverse Kinematics**

Robotics & Actuation + Mechanical Design · Hardware + Software · Week 5

TL-25

Introduction

Industrial robot arms calculate joint angles to reach a point. Students derive the inverse kinematics of a 2-link arm, simulate it in Python, then drive two servos so the arm's pen tip reaches typed coordinates and draws simple shapes. They measure positioning error and improve calibration.

**Software**

Python + Arduino IDE 2

Also: Processing 4

Key hardware (per group)

- Arduino Uno + USB cable x1
- Servo MG995 x2
- 4xAA battery pack x1

Standards

NGSS HS-ETS1 · CSTA Algorithms & Programming (3B)

SDG badges

Activities

GUIDED

- Derive and simulate 2-link inverse kinematics
- Drive the servo arm to target coordinates

DISCOVERY

- Draw a square and measure the error

Outcomes

Maths: Trigonometry and the law of cosines**Engineering:** Robot kinematics and calibration**Computing:** Simulation before hardwareCLASS
26**Robot Swarm: Leader and Follower Cars**

Robotics & Actuation + IoT & Communication · Hardware + Software · Week 5

TL-26

Introduction

Swarms of robots can cooperate on tasks. Students program two 4WD cars: the leader is driven from a phone or a script and broadcasts its moves over ESP-NOW, while the follower copies them and keeps a safe distance using an ultrasonic sensor. They test formation keeping and failure cases.

**Software**

Arduino IDE 2 (ESP32)

Key hardware (per group)

- 4WD robot car (assembled, L298N) x1
- ESP32 Dev Board + USB cable x2
- Ultrasonic sensor HC-SR04 x1

Standards

NGSS HS-ETS1 · CSTA Networks & the Internet (3A)

SDG badges

Activities

GUIDED

- Send leader commands to the follower over ESP-NOW
- Keep a safe following distance with ultrasonic feedback

DISCOVERY

- Add a third car or a formation change

Outcomes

Engineering: Multi-robot coordination**Computing:** Wireless messaging and control loops**Life Skills:** Team engineering

CLASS
27**3D-Printed Gearbox and Gear Ratios**

Mechanical Design • Hardware • Week 6

TL-27

Introduction

Every machine from bicycles to robot arms uses gears to trade speed for force. Students design spur gears in a CAD tool, print a two-stage gearbox, and measure how gear ratio changes the speed and lifting force of a motor. They compare theory with measurements.

**Software**

Onshape or Tinkercad 3D + UltiMaker Cura

Also: BlocksCAD

Key hardware (per group)

- DC hobby motor x1
- 4xAA battery pack x1

Standards

NGSS HS-ETS1 • ISTE Innovative Designer

SDG badges

Activities

GUIDED

- Design and print a two-stage spur gearbox
- Measure speed and lifting force with different ratios

DISCOVERY

- Optimise the ratio to lift a given weight

Outcomes

Engineering: Gear ratios, torque and speed**Maths:** Ratios and measurement**Computing:** Parametric CADCLASS
28**Smart Cane for the Visually Impaired**

Electronics & Circuits + Sensors & Environment • Hardware • Week 6

TL-28

Introduction

Assistive technology helps people live independently. Students design a smart cane that detects obstacles at head and knee height with ultrasonic sensors and warns the user with vibration patterns that get stronger as objects get closer. They interview or research real users and test the cane blindfolded in a safe course.

**Software**

Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- Ultrasonic sensor HC-SR04 x2
- Vibration motor module x1
- Buzzer x1
- 4xAA battery pack x1

Standards

NGSS HS-ETS1 • ISTE Innovative Designer

SDG badges

Activities

GUIDED

- Detect obstacles at two heights
- Map distance to vibration strength and pattern

DISCOVERY

- Test blindfolded and improve with user feedback

Outcomes

Engineering: Assistive device design**Computing:** Sensor fusion and PWM**Life Skills:** Empathy and inclusive design

CLASS
29**Air Quality Data Science for Pakistan**

AI & Data • Software • Week 7

TL-29

Introduction

Smog affects millions of people in Pakistan every winter. Students load open air-quality and weather data for Pakistani cities (for example from OpenAQ or the Open-Meteo air-quality API) into pandas, clean it, plot trends and correlations, and ask questions such as 'Does wind reduce PM2.5?'. They present findings as a short data story with clear charts.

Activities

GUIDED

- Load, clean and summarise the dataset with pandas
- Plot trends and correlations with matplotlib

DISCOVERY

- Write a data story with one recommendation

Outcomes

AI: Data cleaning and exploratory analysis

Science: Air pollution and health

Maths: Correlation and averages

**Software**

Python (pandas, matplotlib) in Google Colab

Also: Jupyter

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Data & Analysis (3B) • NGSS HS-ESS3

SDG badges

Introduction

Behind every AI is a network of simple maths units. Students build a perceptron and then a tiny neural network in Python with NumPy, train it to classify points and learn XOR, and watch the loss go down. They connect it to how tools like Teachable Machine work underneath.

Activities

GUIDED

- Code a perceptron and train it on simple data
- Build a 2-layer network that learns XOR

DISCOVERY

- Visualise the decision boundary as it trains

Outcomes

AI: Neurons, weights, loss and training

Maths: Matrices and gradients (intuition)

Computing: NumPy arrays

**Software**

Python + NumPy (Google Colab)

Also: TensorFlow Playground

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Create) • CSTA AI concepts

SDG badges

CLASS
31**2D Game Development in Godot**

App, Web & Game Development · Software · Week 8

TL-31

Introduction

Godot is a free professional game engine used by indie studios. Students build a 2D top-down game with scenes, nodes, GDScript (a Python-like language), collisions, a UI and sound, then export it to run on Windows or the web. They plan a game design document first.

**Software**

Godot Engine 4

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming (3B) · ISTE Creative Communicator

SDG badges

Activities

GUIDED

- Build a player scene with movement and collisions
- Add enemies, a score UI and sound

DISCOVERY

- Export the game and run a play-test session

Outcomes

Computing: Scenes, nodes, signals and scripting

Arts: Game and level design

Life Skills: Project planning

CLASS
32**Password Security Lab: Hashing and Attacks**

Security & Safety · Software · Week 8

TL-32

Introduction

Websites never store your real password: they store a hash. Students write Python programs to hash passwords, see why short passwords fall to dictionary and brute-force attacks, add salts, and build a password-strength checker. Everything runs on their own test data, with a focus on ethics and the law.

**Software**

Python (hashlib)

Also: picoCTF

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Networks & the Internet (3B) · ISTE Digital Citizen

SDG badges

Activities

GUIDED

- Hash passwords and run a dictionary attack on test hashes
- Add salting and compare attack times

DISCOVERY

- Build a password-strength checker with advice

Outcomes

Computing: Hashing, salts and brute force

Life Skills: Ethical hacking rules and the law

Maths: Combinations and growth

Phase 2 deliberately reuses the existing kit (Arduino Uno, ESP32, ESP32-CAM, sensors, servos, L298N, 18650 cells, solar panels, 3D printer). Only a short list of low-cost parts is new. Suggested order = quantity per group × 6 groups + about 20% spares; adjust to your real group count. Prices to be confirmed with Electrobes.

Status	Part	Max per group	Suggested order (6 groups)	Used in	Notes
HAVE / ORDERED	4WD robot car (assembled, L298N) 4WD chassis, 4 TT motors, L298N driver, battery holder (assembled)	1	7	LE-19, LE-30, YI-25, TL-18, TL-26	Already on the Phase 2B requisition.
HAVE / ORDERED	ESP32-CAM + programmer ESP32-CAM (OV2640) with USB programmer base	1	7	YI-20, TL-18	Already in lab stock.
BUY	WS2812B LED ring (12 LEDs) 12-LED WS2812B (NeoPixel) ring	1	7	LE-21	Keep brightness low when powered from USB.
BUY	WS2812B LED strip (1 m, 30 LEDs) WS2812B 5 V strip, 30 LEDs/m, with leads	1	7	JM-21	Use 100 uF capacitor + 220 ohm data resistor; low brightness on USB.
BUY	MAX7219 8x8 LED matrix MAX7219 8x8 red LED matrix module	1	7	JM-22	
BUY	Passive buzzer Passive piezo buzzer module (plays tones)	1	7	LE-22, JM-23	Different from the active buzzer already in stock.
BUY	Vibration motor module Coin vibration motor module with driver transistor	1	7	YI-22, TL-28	
BUY	Colour sensor TCS34725 TCS34725 RGB colour sensor module (I2C) with white LED	1	7	LE-23	
BUY	Load cell 5 kg + HX711 5 kg bar load cell with HX711 amplifier board	1	7	JM-26	Needs a simple card or acrylic platform.
BUY	Current sensor INA219 INA219 I2C current/voltage/power sensor	1	7	YI-27	
BUY	Wooden skewers Bamboo skewers (axles)	4	28	LE-28, JM-28, YI-26, TL-27	
BUY	Split pins (paper fasteners) Brass split pins for card linkages	12	86	YI-26	

Teacher checks

- Test every new module once before class (LED ring, matrix, colour sensor, load cell, INA219).
- WS2812B strips: keep brightness at or below about 20% on USB power, or use a separate 5 V supply.
- Robot sumo (YI-25) and swarm (TL-26) need cars from the Phase 2B requisition; TL-26 shares 2 cars between 2 groups.
- Software-only classes need just a laptop per group (or per student where available).

After classes 17-32, two optional extra sessions turn skills into a team project. They sit outside the 16-class batch and teachers plan them separately; this page sets the shared structure so every pathway ends with a public product.



Session	Focus	Output
Extra A	Define the problem and user, plan the design, choose parts from past kits, start building	Project plan + first prototype
Extra B	Test, get feedback, improve, rehearse and present at the showcase	Working prototype + 2-minute presentation

Suggested projects by pathway

Little Explorers • Ages 5-7

- A robot helper for our classroom
- A light-and-sound greeting machine
- An interactive Scratch story with a real button
- A cardboard automaton that tells a story

Junior Makers • Ages 8-11

- A smart lunchbox scale that tracks food waste
- A Python or MakeCode game for younger students
- A light-show costume or poster
- A smart crossing for the school gate

Young Innovators • Ages 12-14

- A data-logging study room monitor with charts
- An accessible game controller for a classmate
- A wildlife camera survey of the school garden
- An AI tool that helps someone communicate

Tech Leaders • Ages 15-17

- A vision-guided sorting robot
- A smog awareness dashboard for our city
- An assistive device tested with a real user
- A complete Godot game released to the school

Simple assessment rubric (1-4)

Criterion	What we look for
Problem & user	Clear problem and who it helps
Build & function	Prototype works; wiring and code are tidy
Process	Evidence of testing, feedback and improvement
Communication	Explains how it works and what they learned
Teamwork	Shared roles and respectful collaboration